Smart Home Technology Solutions for Individuals with Intellectual and Developmental Disabilities

Enhancing Independent Living Through Assistive Technology and Reducing Caregiver Burden

Olzhas Yessenbayev

Cornell Tech
Center for Family Support Partnership
Technology Assessment Report

New York, July 2025

Contents

1 Introduction

Too often, independence is something individuals with **intellectual and developmental disabilities (IDD)** strive for but find elusive. IDD encompasses a range of lifelong conditions that affect cognitive abilities, adaptive skills, and social interactions—making tasks like learning routines, managing emotions, or handling household chores feel like steep climbs. For many, daily life brings hurdles such as preparing a meal without mishaps, remembering medications amid distractions, or navigating a home safely without constant guidance. These challenges not only limit personal freedom but can lead to crises—unattended stoves sparking fires, nighttime elopements causing distress, or isolation amplifying health inequities. Life with IDD therefore becomes constrained, dependent on constant oversight by others. To foster autonomy and well-being, nothing matters more than breaking these barriers.

Promoting resident independence has always been a core part of CFS's mission. The Center for Family Support (CFS) is working to change how individuals with IDD, their families, and caregivers approach independence. CFS began as a pioneering force against institutionalization, advocating for community-based alternatives. Over seven decades, CFS has grown into a multi-state non-profit serving over 1,050 individuals with IDD and their families across the greater New York City area and Eastern New Jersey, with around 1,500 staff members. Its person-centered services range from residential programs and day habilitation to supported employment, respite care, and self-directed options. At the heart of CFS work are the residential homes—approximately 250 group homes, community homes, and congregate living arrangements as of Fiscal Year 2024—where CFS supports residents in safe, inclusive environments tailored to their needs. Through Individualized Service Plans (ISPs), CFS has emphasized skill-building, community integration, and self-advocacy, as seen in stories of many residents, who transitioned from CFS homes to independent living, balancing family, employment, and community involvement with the support of CFS.

Smart home technology is the next step in CFS's efforts to promote inclusivity, advocacy, and collaboration to shape technology for public good. Imagine a home that acts as a silent guardian: a central hub connecting sensors to detect wandering or falls, interactive tablets for friendly reminders and video calls, and automated systems for medication management or kitchen safety. For residents, this means greater autonomy and independence—fewer forgotten tasks leading to emergencies, and more confidence in self-care. For caregivers—whether family or Direct Support Professionals—it means more time for meaningful interactions and support, and the potential for reduced overtime through fewer incidents.

This report unfolds in several key sections. We begin with a foundational *Background* on the landscape of IDD and the support structures in place. Following this, the *Technology Overview* details the specific smart home components and their architecture. A deeper dive into *Specific*

Sensing Technologies evaluates the advantages and disadvantages of various market-available devices. The *Implementation Guide* provides a practical roadmap for integrating these technologies within CFS homes. Finally, we conclude by summarizing the key takeaways and outlining future directions.

2
Background

Author: Olzhas Yessenbayev

2.1 People with Intellectual and Developmental Disabilities (IDD)

2.1.1 Causes, Conditions, and Prevalence

Intellectual and developmental disabilities (IDD) refer to a diverse group of lifelong conditions that originate during the developmental period—typically before birth or in early childhood—and are characterized by significant limitations in both intellectual functioning (such as learning, reasoning, and problem-solving) and adaptive behaviors (including conceptual, social, and practical skills needed for everyday life). These disabilities affect how individuals interact with their environment, often impacting physical, cognitive, language, or behavioral abilities, and they persist throughout a person's lifetime. Common examples include autism spectrum disorder, cerebral palsy, Down syndrome, fragile X syndrome, fetal alcohol spectrum disorders, and intellectual disability itself, each varying in severity and presentation.

> **Note**
>
> Globally, an estimated 1.3 billion people—or about 16% of the population—experience significant disabilities, including IDD, with prevalence influenced by factors like increasing noncommunicable diseases and longer life expectancies. In the United States, recent estimates show that about 1 in 6 children aged 3–17 (around 17%) have one or more developmental disabilities, occurring across all racial, ethnic, and socioeconomic groups.

The range of potential factors contributing to IDD is multifaceted, often involving a complex interplay of genetic, environmental, and other influences. Genetic causes include mutations, additions, or deletions in genes (e.g., changes in the MECP2 gene leading to Rett syndrome) and chromosomal abnormalities like an extra chromosome 21 in Down syndrome. Environmental factors encompass prenatal exposures, such as alcohol (causing fetal alcohol spectrum disorders), infections like cytomegalovirus during pregnancy, or toxins like lead; complications during birth, such as oxygen deprivation; and postnatal issues like traumatic brain injury or untreated newborn jaundice leading to kernicterus. Other risk factors include preterm or low birthweight, multiple births, maternal health behaviors (e.g., smoking or drinking during pregnancy), and broader social determinants like poverty or lack of access to healthcare. While some causes are identifiable, many IDD cases result from unknown or combined factors, highlighting the need for ongoing research.

2.1.2 Limitations in Daily Life

Individuals with IDD face substantial challenges in daily life, which can vary widely based on the disability's type and severity but often revolve around limitations in adaptive functioning, communication, self-care, and social interactions. Common difficulties include learning new skills, managing personal hygiene, preparing meals, navigating public transportation, or maintaining employment, which can lead to unsafe conditions, reduced quality of life, and social isolation. Health inequities exacerbate these issues; people with IDD are twice as likely to develop comorbid conditions like depression, diabetes, obesity, asthma, or poor oral health, and they face barriers such as stigma, discrimination, inaccessible healthcare facilities, and exclusion from education or work opportunities. During crises like the COVID-19 pandemic, those with IDD experienced higher mortality rates and neglect in institutional settings, underscoring vulnerabilities in emergency responses. These challenges not only affect personal autonomy but also contribute to poorer overall health outcomes, with individuals often dying earlier than their non-disabled peers.

2.1.3 Need for Support

Reliance on caregivers is a hallmark of life with IDD, as many individuals require ongoing support for daily activities, decision-making, and community participation. Family members, particularly women and girls, often serve as primary caregivers due to gaps in formal support systems, facing high levels of emotional, physical, and financial burden that can lead to mental health issues like anxiety, depression, and burnout. Caregivers report constant mental pressure from managing antisocial behaviors, medical needs, and long-term planning, with many experiencing economic strain from reduced work opportunities. Over 70% of adults with IDD in the U.S. live with family, highlighting this dependency, though professional supports like direct support professionals can alleviate some pressures when available.

Living conditions for people with IDD are frequently marked by poverty, exclusion, and inadequate accommodations, influenced by societal barriers and limited access to resources. Many reside in family homes, group homes, or community-based settings, but institutionalization persists in some cases, leading to isolation and reduced quality of life. Challenges include inaccessible transportation (15 times more difficult for those with disabilities), poor housing quality, and exclusion from education or employment, increasing risks of violence (e.g., women with disabilities are 2–4 times more likely to face intimate partner abuse) and unmet health needs. Globally, frameworks like the UN Convention on the Rights of Persons with Disabilities advocate for better inclusion, but inequities remain, emphasizing the need for supportive environments that promote independence and community integration.

2.2 Support Structures

2.2.1 Overview

Support structures for individuals with IDD encompass a range of services aimed at promoting independence, community inclusion, and quality of life. These include residential programs, day habilitation, supported employment, respite care, and self-directed services, often funded through Medicaid and state programs.

2.2.2 Support Homes

Support homes, such as those operated by organizations like the Center for Family Support (CFS), provide safe, community-integrated environments tailored to IDD needs, fostering independence while reducing reliance on large institutions. CFS manages group homes, community homes, and congregate living options, housing approximately 250 individuals across New York and New Jersey as of Fiscal Year 2024. Group homes typically accommodate 3-6 residents with on-site support for skill-building, while community options offer flexibility and neighborhood integration. Features include accessibility modifications, sensory-friendly designs, and self-directed elements like Individualized Service Plans (ISPs). Funding comes from Medicaid, grants, and donations, with eligibility requiring age 18+, an approved ISP, and disability criteria.
These homes have profound impacts, empowering residents through skill development and social engagement. For example, a CFS resident transitioned from group homes to independent living, balancing motherhood, employment, and community involvement, crediting programs like self-defense classes for her confidence. Overall, such homes enhance dignity, reduce isolation, and support long-term outcomes like sustained employment and resilience.

2.2.3 Direct Support Professionals (DSPs) and Their Range of Responsibilities

Direct Support Professionals (DSPs) play a multifaceted role in supporting individuals with intellectual and developmental disabilities (IDD), extending far beyond basic caregiving to encompass empowerment, skill development, and community integration. Their responsibilities are highly individualized, adapting to each person's unique needs, preferences, and goals, often outlined in Individualized Service Plans (ISPs). Core duties include assisting with activities of daily living (ADLs) such as personal hygiene, grooming, meal preparation, medication administration, and household chores, while promoting independence by encouraging self-performance whenever possible. DSPs also teach adaptive life skills, including financial management (e.g., budgeting and banking), using public transportation, grocery shopping, time management, and personal scheduling, to foster greater autonomy in everyday life.
Health and wellness support forms another critical layer, where DSPs monitor physical, behavioral, and mental health changes; accompany individuals to medical appointments; promote healthy lifestyles through exercise, nutrition planning, and habit-building; and offer emotional support for conditions like anxiety or depression. They facilitate social and recreational engagement by organizing community outings, supporting participation in hobbies (e.g., art, music, or sports classes), building relationships to reduce isolation, and encouraging volunteering or civic involvement. In vocational realms, DSPs provide job coaching, resume assistance, workplace accommodations, and ongoing employment support to help individuals secure and maintain meaningful work.
Advocacy is a cornerstone, with DSPs collaborating with families, healthcare providers, educators, and service coordinators to ensure holistic care; responding to crises; documenting progress; and teaching self-advocacy skills. This vast scope often requires DSPs to juggle roles akin to clinicians, administrators, and mentors, working in diverse settings like homes, day programs, or workplaces, sometimes with 24/7 availability. Research highlights how this comprehensive support directly correlates with improved quality of life outcomes, such as enhanced independence, better health, and stronger social connections for people with IDD.

2.2.4 Practical Difficulties and Burdens Faced by DSPs

The breadth of DSP responsibilities, while rewarding, contributes to significant practical challenges, creating a workforce crisis characterized by high turnover, burnout, and systemic barriers. Annual turnover rates often exceed 40-50% in IDD support programs, driven by low wages (typically $15-18 per hour), inadequate benefits, and demanding workloads that include long shifts, overtime, and multiple jobs to make ends meet. Emotional and physical burdens are profound: DSPs frequently experience stress, trauma, and burnout from constant monitoring, crisis intervention, and exposure to verbal, physical, or emotional abuse from clients, compounded by their own adverse childhood experiences (e.g., higher rates of emotional abuse or household mental illness).

Recruitment and retention issues are exacerbated by insufficient training, autonomous decision-making in high-stakes situations, and a lack of professional recognition, leading to vacancies that strain remaining staff and compromise care quality. For instance, DSPs supporting aging individuals with IDD face additional barriers like managing superimposed dementia, navigating inaccessible services, and addressing societal stigma, which heighten workload demands. These challenges have ripple effects: high turnover correlates with increased emergency department visits, incidents of abuse/neglect, and injuries for people with IDD, while agencies incur substantial costs for training and recruitment, estimated in the billions annually across the U.S. Qualitative studies reveal DSPs' perceptions of role overload, with many reporting workplace stress and emotional exhaustion that impact care consistency. Diversity disparities in the workforce, including underrepresentation and unequal support for minority DSPs, further deepen the crisis.

Opportunities for Improvement Through Technology

Technology presents promising avenues to mitigate these burdens, enhancing DSP efficiency, reducing oversight needs, and promoting independence for individuals with IDD without replacing human support. Smart home systems and sensors (e.g., for activity monitoring, fall detection, or kitchen safety) can provide real-time alerts, automate routine checks, and enable remote supervision, allowing DSPs to focus on higher-value interactions while cutting costs and preserving dignity. Communication devices like tablets, voice-activated assistants (e.g., smart speakers), and apps for visual schedules or augmentative and alternative communication (AAC) support diverse modes of interaction, helping DSPs teach skills and reduce isolation.

AI-powered tools streamline administrative tasks: electronic health records (EHR) systems facilitate data tracking, progress documentation, and billing, while AI for clinical workflows and time-stamped notes eases end-of-shift reporting and reduces burnout. Remote support technologies, including video monitoring and computer vision AI for procedure automation, enable scalable independence in living and work environments, addressing workforce shortages by supplementing DSP availability. Emerging trends like digital literacy initiatives and assistive wearables (e.g., health-tracking watches) further empower users, with studies showing potential for broader adoption in IDD services to improve outcomes ethically and collaboratively. Overall, these innovations align with goals of augmenting DSP roles, potentially lowering turnover by making work more sustainable and impactful.

CFS Group homes

3 Technology Overview

Author: Olzhas Yessenbayev

3.1 Vision of the Smart Home for IDD Residents

This section describes the architecture of the smart home system designed to support individuals with intellectual and developmental disabilities (IDD) in community-based residential settings, such as those provided by Community Family Services (CFS). The system aims to enhance safety, independence, and quality of life through non-intrusive monitoring and targeted interventions.

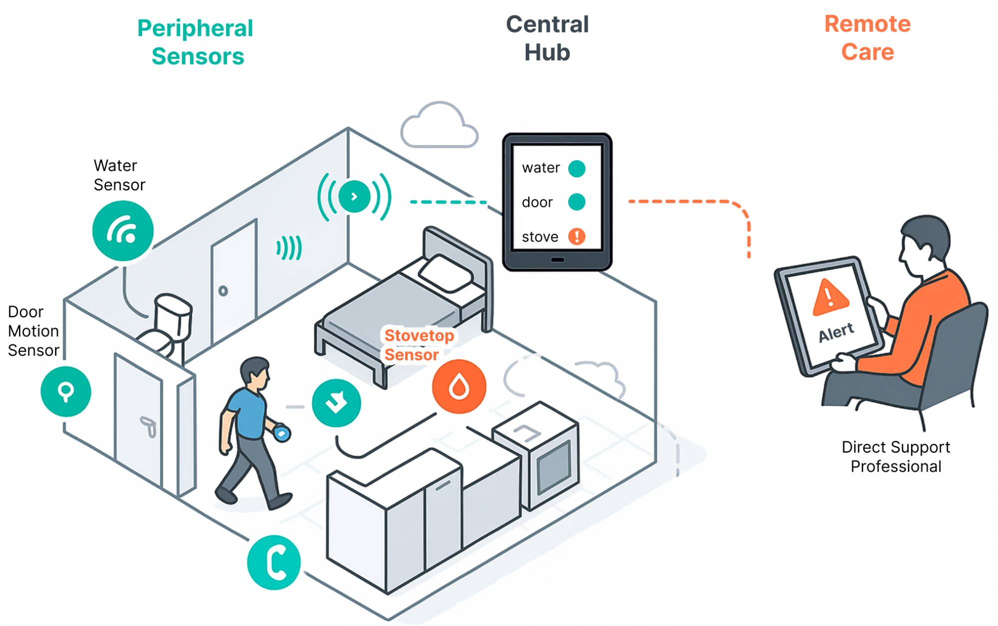

Figure 3.1: *System overview: Smart home technology integration for IDD support in CFS residences.*

The system comprises three main components: peripheral sensors, a central hub, and remote care access. These elements integrate to monitor daily activities, detect potential risks, and facilitate support from direct support professionals (DSPs). Residents in these settings typically live in shared or individual apartments within supervised facilities, where the system is installed to cover key areas like bedrooms, kitchens, bathrooms, and common spaces. The residential space is adapted to accommodate sensors without altering the home's layout significantly, ensuring compatibility with existing infrastructure.

Scenario

Imagine a resident preparing a meal in the kitchen: the stovetop sensor continuously collects data, sending usage information to the central hub. The hub records and analyzes this data, learning typical cooking patterns. When the stovetop is left unattended for 45 minutes—beyond the usual 15-20 minute sessions—the hub alerts the DSP professional via mobile app. The DSP can then call with a reminder, visit to assist, or activate an audio prompt remotely. This shifts care from constant supervision to targeted, as-needed support while preserving the resident's independence.

3.1.1 Three Key Components

The system's framework involves sensors for data collection, a hub for processing and analysis, and remote access for caregiver intervention. Data is transmitted wirelessly from sensors to the hub using protocols such as Bluetooth Low Energy (BLE) or Wi-Fi. The hub applies predefined rules and machine learning algorithms to identify anomalies. Alerts are then securely forwarded to authorized users via cloud-based platforms, ensuring compliance with privacy standards like HIPAA in the U.S. This setup allows for customization based on individual resident needs, such as setting thresholds for activity levels or environmental conditions.

3.1.2 Peripheral Sensors: Capturing Everyday Moments

Peripheral sensors are deployed throughout the residential space to monitor activities and environmental conditions relevant to IDD residents. These battery-powered or plugged-in devices include motion sensors (e.g., PIR sensors detecting movement in rooms), door/window contact sensors (magnet-based to track openings), water leak detectors (moisture-sensitive probes in bathrooms and kitchens), stovetop sensors (temperature and vibration monitors), and wearable devices (optional, like smartwatches for heart rate or fall detection). Placement is strategic: motion sensors in hallways and living areas, door sensors on exits to prevent wandering, and water sensors under sinks or near toilets to detect floods.
For IDD residents, who may face challenges with memory, executive function, or sensory processing, these sensors provide data on routine adherence (e.g., medication cabinet access) and risk factors (e.g., prolonged inactivity indicating a fall). Sensors operate passively, with low power consumption and minimal maintenance, typically lasting 1-2 years on batteries. Data is anonymized at the source where possible to protect privacy.

Wandering Detection

Door sensors monitor exit patterns, flagging openings outside normal hours (e.g., nighttime) to alert for potential wandering, a common concern in IDD populations.

Safety Prevention

Water sensors detect overflows in bathrooms, notifying staff to mitigate slip or flood risks promptly.

Additional sensor types may include environmental monitors for temperature/humidity (to prevent heat-related issues) and smart plugs for appliance usage tracking. Integration ensures comprehensive coverage without overwhelming the space.

3.1.3 Central Hub: The Heart of Coordination

The central hub serves as the processing unit, aggregating data from sensors via wireless connections. Typically a wall-mounted tablet or dedicated device (e.g., models like Amazon Echo Show or custom Raspberry Pi-based systems), it runs software that applies rules-based logic and basic AI to analyze patterns. For instance, if motion sensors indicate no activity in the bedroom for an extended period during expected sleep times, the hub could flag a potential issue like insomnia or a medical event.

The hub stores data locally with cloud syncing for redundancy, using encryption to secure information. Residents can interact with the hub through a simplified interface displaying status icons (e.g., green for normal, red for alerts) for doors, water, or appliances. DSPs configure the hub during setup to align with resident profiles, including thresholds for alerts based on historical data.

In residential spaces, the hub is placed in a central location like the living room for easy access, supporting voice commands or touch inputs for residents with varying abilities.

3.1.4 Remote Care Access: Extending Support Beyond the Home

Remote care access enables DSPs, family members, or other caregivers to monitor and respond via secure mobile apps or web portals. DSPs, who provide hands-on support in IDD settings, access the system through authenticated accounts on platforms like iOS/Android apps (e.g., similar to those from providers like Caregiver or custom solutions from companies like Essence Group or Tunstall). Alerts are pushed as notifications with details (e.g., "Door open for 30 minutes at 2 AM"), allowing DSPs to view live sensor data, historical logs, or even activate responses like automated lights or voice messages.

Access is role-based: DSPs have full intervention capabilities, while family members might have view-only permissions. Security features include two-factor authentication, end-to-end encryption, and audit logs. In cases like detected wandering, a DSP can use GPS integration (if enabled) to locate the resident or coordinate with on-site staff.

This component reduces the need for constant physical presence, allowing DSPs to manage multiple residents efficiently while respecting privacy through consent-based data sharing.

Together, these components form an integrated system that supports IDD residents in maintaining independence within safe parameters.

4

Specific sensing technologies

Author: Olzhas Yessenbayev

In this section, we overview a range of assistive technologies designed for smart home integration to support individuals with Intellectual and Developmental Disabilities (IDD). Drawing on both real-world implementations and academic research, we highlight the broad categories of technology that address key needs—such as safety, independence, and communication—while also supporting Direct Support Professionals (DSPs) in their roles. We attempt to outline the practical impact of these technologies, including measurable benefits like cost savings, reduced incidents, and improved quality of life for residents:

- **Cost**: Estimated initial and ongoing expenses, with attention to scalability and affordability.
- **Ease of Integration**: How readily the technology can be installed and connected to existing smart home systems.
- **Reliability**: Real-world performance, including uptime and the likelihood of false alarms.
- **IDD-Specific Applicability**: The extent to which the technology addresses cognitive, sensory, or behavioral needs typical in IDD populations, such as through simplified interfaces or customizable alerts.

4.1 Technology Overview

Table 4.1: *Simply Home Technology Overview for IDD Support*

Device/Sensor	Description & Functionality	Cost (Purchase + Monthly Fee)	IDD Usage Example	Linked Outcomes
Door Sensor	Wireless contact sensor that detects door opening/closing. Triggers alerts for unusual activity (e.g., wandering at night). Integrates with hub for custom rules like "If front door opens after 9 PM, alert caregiver and play verbal prompt: 'It's bedtime, please close the door.'"	$50-100 (separate); Included in bundles. Monthly: Part of system fee ($79.95-$149.95).	In a CFS home, installed on exterior doors. Resident with IDD wanders; sensor alerts staff app. Staff remotely locks door via app or prompts via tablet speaker, preventing elopement without physical intervention. Data logged for trends (e.g., 20% reduction in nighttime exits).	Impact: Enhances safety and independence. Efficiency: Reduces overnight staffing by 15-20 hours/week (~$300-400 savings). Behavioral Change: Fewer wandering incidents, tracked via portal reports.

Continued on next page...

Table 4.1: *Simply Home Technology Overview for IDD Support (continued)*

Device/Sensor	Description & Functionality	Cost (Purchase + Monthly Fee)	IDD Usage Example	Linked Outcomes
Motion Sensor	Detects movement in rooms or areas. Can trigger lights, alerts for inactivity (e.g., falls), or routines (e.g., no motion in kitchen after 30 min = stove check).	$40-80 (separate); Included in bundles. Monthly: Part of system fee.	Placed in living room/bedroom. For IDD resident with fall risk, no motion for 10 min triggers alert to caregiver's phone; caregiver views via integrated Ring camera and responds with 2-way audio on home tablet, avoiding unnecessary visits.	Impact: Promotes safe activity monitoring. Efficiency: Cuts response time by 50%, saving ~$200/month in emergency calls. Behavioral Change: Increased activity levels, measured by logged patterns (e.g., 10% more daily movement).
Panic Pendant	Wearable button (necklace/wrist) for emergency alerts. Waterproof, with optional fall detection. Press activates 24/7 Response Center or custom responders.	$50-150 (separate, incl. fall-detect: $144.95-$284.95). Monthly: $27.95-$51.95.	IDD resident wears it; presses for help (e.g., during seizure). Alert goes to family/staff app; Response Center calls emergency if needed. In CFS homes, data shows reduced response delays.	Impact: Immediate assistance fosters confidence. Efficiency: Replaces constant supervision, saving $500/month per resident. Behavioral Change: More self-initiated help-seeking, with logs showing 25% fewer unreported incidents.
Bed/Chair Sensor (Pressure Pad)	Pad under mattress/chair detects getting in/out. Alerts for prolonged absence (e.g., night wandering) or falls.	$100-200 (separate); Included in bundles. Monthly: Part of system fee.	Under bed in CFS shared room. Resident leaves bed at 3 AM; sensor alerts staff remotely, who issues prompt via hub speaker ("Time to sleep") or checks via app, reducing need for physical checks. Trends show improved sleep patterns.	Impact: Better rest and health. Efficiency: Eliminates overnight staff presence, saving $1,000/month. Behavioral Change: 15-20% improvement in sleep adherence, quantified by portal data.
Stove Sensor	Detects if electric/induction stove is on/off (not heat). Triggers alerts for prolonged use (e.g., >1 hour = fire risk). Does not work on gas stoves.	$150-250 (separate); Included in bundles. Monthly: Part of system fee.	Installed on kitchen stove. IDD resident forgets stove; sensor alerts caregiver app. Caregiver remotely turns off via outlet switch (if integrated) or prompts via tablet, preventing accidents. Logs track cooking safety trends.	Impact: Reduces burn/fire risks. Efficiency: Lowers insurance claims by ~$500/year. Behavioral Change: Safer cooking habits, with 30% fewer forgotten appliances per logs.
Water Sensor	Detects leaks/floods (e.g., forgotten faucet). Triggers alerts but no automatic shut-off.	$50-100 (separate); Included in bundles. Monthly: Part of system fee.	Placed near sink/shower. Water overflow alerts staff; they respond remotely via app to guide resident or dispatch help, avoiding damage.	Impact: Prevents water-related hazards. Efficiency: Saves $200-500 in repair costs per incident. Behavioral Change: Improved hygiene routines, tracked by fewer alerts over time.
Smart Thermostat	Controls temperature remotely. Integrates with hub for schedules (e.g., lower at night).	$100-200 (separate); Included in Environmental Controls bundle. Monthly: Part of system fee.	Resident's tablet/app adjusts temp; auto-schedules prevent discomfort. Staff monitors via portal for health (e.g., no overheating).	Impact: Comfortable living environment. Efficiency: Energy savings of 10-20% (~$50/month). Behavioral Change: Resident learns routine adjustments, reducing complaints by 15%.
Automated Door Lock-/Opener	Remote locking/unlocking via app/tablet. Integrates with Ring for visitor verification.	$150-300 (separate); Included in Environmental Controls. Monthly: Part of system fee.	Front door: Visitor rings (via Ring); resident/staff unlocks from tablet. For IDD, staff manages to prevent unauthorized entry.	Impact: Secure access control. Efficiency: Reduces key management, saving staff time (~5 hours/week). Behavioral Change: Safer visitor interactions, with logs showing fewer unannounced entries.
Outlet Switch/Light Switch	Remote on/off for appliances/-lights. Mechanical faceplate for lights.	$50-150 (separate); Included in bundles. Monthly: Part of system fee.	Lamp/plug: Auto-off at bedtime via schedule. Resident controls via tablet switch for independence.	Impact: Energy/safety management. Efficiency: Lowers utility bills by 15%. Behavioral Change: Consistent routines, e.g., lights off promoting better sleep.
Medication Dispenser	Automated tray dispenses up to 4 doses/day with alerts (buzzer/light). Locks to prevent errors.	$219.95-$299.95 + $24.95-$29.95/mo.	Resident hears buzzer; misses dose = alert to staff app. Pharmacy pre-fills trays. Reduces re-hospitalizations by ensuring compliance.	Impact: Healthier medication habits. Efficiency: Cuts nursing visits by 50% (~$400/month). Behavioral Change: 80-90% adherence rate, per studies and logs.
Ring Video Doorbell	Camera/doorbell with motion detection, 2-way talk, HD video. Integrates with hub.	$251.95-$329.95 (optional add-ons $30/year for recording).	Door motion: Alert to app; staff views/speaks to visitor, unlocking if safe. For IDD, prevents stranger entry.	Impact: Enhanced security/privacy. Efficiency: Replaces in-person checks, saving $300/month. Behavioral Change: Residents feel safer, with fewer anxiety incidents logged.

4.2 Health

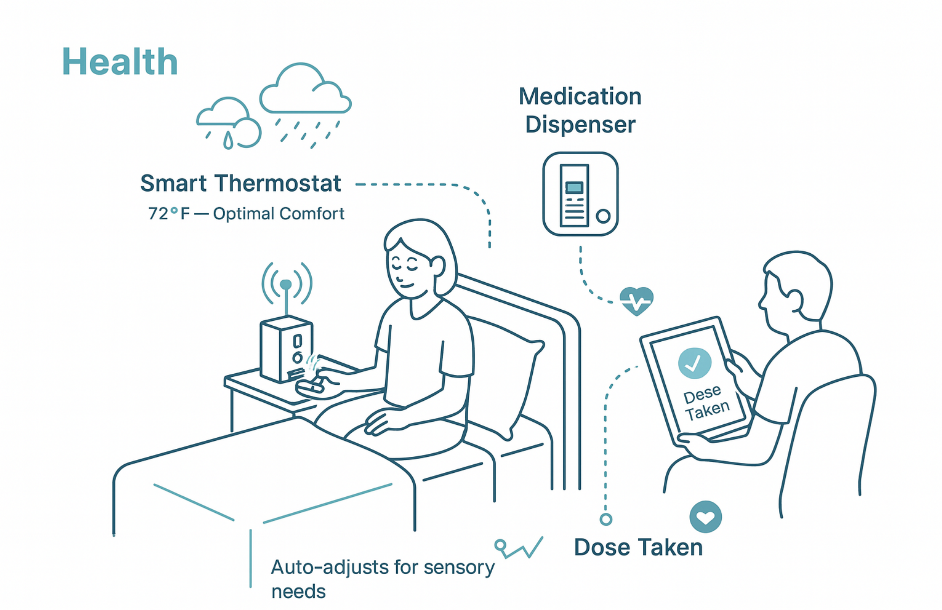

Figure 4.1: *Examples of health-related sensors and devices in a smart home for IDD support.*

Assistive technologies in the health domain focus on monitoring and managing physical well-being, medication adherence, and environmental factors that influence daily health for individuals with intellectual and developmental disabilities (IDD). These tools, such as smart thermostats and medication dispensers, provide automated reminders, environmental adjustments, and remote oversight to prevent health complications and promote self-management. Individuals with IDD often struggle in this area due to cognitive challenges like memory deficits, difficulty recognizing bodily cues (e.g., discomfort from temperature extremes), or executive functioning issues that lead to inconsistent medication routines, resulting in higher rates of hospitalizations, exacerbated chronic conditions, and increased dependency on caregivers.

4.2.1 Smart Thermostat

Functionality: Smart thermostats are programmable devices that automatically regulate home heating, cooling, and ventilation systems by learning user preferences, schedules, and environmental conditions through built-in sensors for temperature, humidity, occupancy, and sometimes air quality. They react to inputs like manual adjustments via a touchscreen interface, voice commands through integrated assistants (e.g., Alexa or Google Assistant), or app controls on smartphones, and can auto-adjust based on external factors such as weather data from the internet, geofencing (detecting when residents leave or return home), or motion detection to optimize energy use— for instance, lowering the temperature when no one is home and raising it upon return. Typically placed on a wall in a central location like a hallway or living room, replacing an existing thermostat, they connect wirelessly to the home's Wi-Fi and HVAC system for seamless integration with other smart devices, allowing reactions like triggering alerts if temperatures deviate from safe ranges or linking with sensors to maintain consistent comfort levels.
How it helps IDD: Individuals with intellectual and developmental disabilities (IDD) often face challenges in managing environmental comfort due to difficulties with cognitive processing, memory, or fine motor skills, which can lead to forgetting to adjust thermostats, resulting in uncomfortable living conditions, health risks like overheating or hypothermia, increased energy

costs, and reliance on caregivers for basic temperature control—exacerbating feelings of dependency and potentially leading to behavioral issues from discomfort. A smart thermostat addresses these by automating temperature regulation based on learned routines or preset schedules tailored to the resident's needs, such as maintaining warmer settings during mornings for those with sensory sensitivities; it enables voice or app-based control to bypass physical manipulation, sends alerts to caregivers if unsafe temperatures are detected (e.g., via integrated air quality monitoring for respiratory issues common in IDD), and promotes independence by allowing residents to use simple commands like "Alexa, set temperature to 72 degrees," reducing caregiver interventions by up to 20-30% according to studies on assistive tech in supported living, while also tracking usage patterns to inform personalized care plans and prevent health complications.

Various vendors (together with cost): Leading options include the Ecobee Smart Thermostat Premium ($250, ecobee.com, with remote sensors and voice assistant integration), Google Nest Learning Thermostat ($249, store.google.com, known for adaptive learning), Honeywell Home T9 ($169, honeywellhome.com, with room sensors), and Sensi Touch 2 ($149, sensi.emerson.com, large touchscreen). All support app/voice control; some require professional installation.

4.2.2 Medication Dispenser

Functionality: Medication dispensers are automated devices that store, organize, and dispense pre-sorted pills or medications at scheduled times, typically holding multiple doses in locked compartments that open via timers, with features like audible alarms, flashing lights, or voice reminders to prompt intake; they react to user interactions such as button presses to release doses early (if allowed) or missed doses by sending notifications to connected apps or caregivers, and some integrate with sensors to detect if medication was removed or use AI to track adherence patterns. Placed on a countertop, nightstand, or wall-mounted in accessible areas like kitchens or bedrooms, they connect to Wi-Fi or cellular networks for remote monitoring, app-based programming of schedules by caregivers, and integration with smart-home ecosystems to trigger alerts or link with voice assistants for reminders like "Time for your medicine."

How it helps IDD: Residents with IDD frequently struggle with medication management due to cognitive impairments affecting memory, time perception, or executive functioning, leading to missed doses, overdoses, or inconsistent adherence that can result in health deterioration, hospitalizations (up to 2-3 times higher rates in IDD populations per CDC data), increased caregiver burden for supervision, and reduced independence in daily self-care routines. A medication dispenser mitigates these issues by automating dispensing with locked, tamper-proof compartments to prevent errors, providing multi-sensory cues (sounds, lights, voices) tailored to sensory preferences common in IDD like autism spectrum disorders, and alerting caregivers via apps or calls for non-compliance—enabling remote oversight that reduces in-person checks by 40-50% based on trials in developmental disability support programs; this fosters self-management skills, improves adherence rates to 85-95% as shown in studies from the Journal of Intellectual Disability Research, and empowers residents with simple interfaces or voice controls, ultimately enhancing dignity, health outcomes, and reducing emergency interventions.

Various vendors (together with cost): Notable vendors include Hero Health ($99 setup + $29.99/mo, herohealth.com, up to 10 meds, pharmacy integration), MedMinder ($49.99/mo, medminder.com, locked trays, voice reminders), Philips Lifeline ($59.95/mo, lifeline.philips.com, up to 60 doses, emergency response), Livi ($199 + $19.99/mo, mylivi.com, customizable alerts), and e-pill MedSmart PLUS ($299, epill.com, no subscription, basic use).

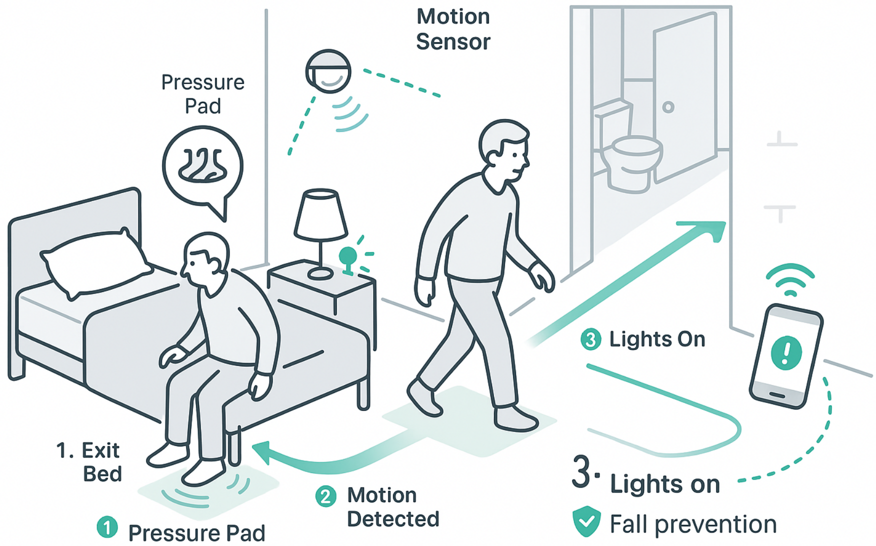

Figure 4.2: *Examples of mobility-related sensors and devices in a smart home for IDD support.*

4.3 Mobility

Mobility-related assistive technologies aim to support safe movement and activity tracking within the home, using sensors to detect patterns, prevent falls, and encourage independence for those with IDD. Devices like motion sensors and pressure pads monitor occupancy and transitions, alerting caregivers to anomalies while fostering autonomous navigation. IDD individuals frequently face mobility struggles, including wandering behaviors driven by sensory overload or disorientation, heightened fall risks from motor impairments or poor spatial awareness, and sedentary patterns that contribute to physical decline and isolation, often necessitating constant supervision that limits personal freedom.

4.3.1 Motion Sensor

Functionality: Motion sensors, often utilizing passive infrared (PIR) technology or microwave radar, detect movement within a specified range by sensing changes in heat signatures or motion patterns, typically covering areas up to 30-40 feet with a 90-120 degree field of view. When motion is detected, the sensor reacts by sending wireless signals to a connected hub or app, triggering customizable actions such as activating lights, sounding alerts, or notifying caregivers via smartphone notifications or integrated systems; they can be programmed for sensitivity adjustments to ignore pets or small movements and often include tamper detection for security. These sensors are commonly placed in high-traffic areas like living rooms, hallways, bedrooms, or entryways, mounted on walls, ceilings, or corners at heights of 6-8 feet to optimize coverage while avoiding direct sunlight or heat sources that could cause false triggers.
How it helps IDD: Individuals with intellectual and developmental disabilities (IDD) often face challenges such as wandering behaviors, especially at night, increased risk of falls due to impaired coordination or judgment, and periods of prolonged inactivity that could indicate health issues like seizures or depression, leading to safety concerns and higher dependency on constant caregiver supervision. Motion sensors address these by providing non-intrusive monitoring of activity patterns, alerting caregivers to unusual movements (e.g., nighttime wandering) or lack thereof (e.g., no motion for extended periods suggesting a fall), enabling timely remote interventions like voice prompts through integrated speakers or app-based checks, which promotes

greater independence, reduces the need for 24/7 staffing, and can lead to behavioral improvements through data-driven insights on routines.

Various vendors (together with cost): Options include Aqara Motion Sensor P1 ($25, amazon.com, hub required), Simply Home ($40-80, simply-home.com, IDD-focused bundles), TP-Link Kasa KS105 ($20, tp-link.com, no hub), and Samsung SmartThings ($25, samsung.com, wide integration).

4.3.2 Bed/Chair Sensor (Pressure Pad)

Functionality: Bed or chair sensors, typically thin pressure-sensitive pads placed under a mattress, cushion, or sheet, detect changes in weight or pressure to monitor occupancy, reacting by sending alerts when someone gets in or out; they use wired or wireless connections to a hub, triggering notifications for prolonged absence (e.g., after 10-30 minutes) or immediate exits, with adjustable sensitivity to avoid false alarms from minor shifts, and some models include vibration or fall detection features. These pads are positioned directly on the bed frame under the mattress for beds or on the seat/base for chairs, ensuring full coverage of the surface.

How it helps IDD: Residents with IDD commonly experience sleep disturbances, nighttime wandering, or risks of falling out of bed/chair, which can result in injuries, disrupted routines, and the necessity for constant overnight monitoring by caregivers, increasing burden and costs while limiting resident autonomy. Pressure pad sensors ease these issues by providing automated alerts for bed/chair exits, allowing caregivers to respond remotely with voice reminders or checks via integrated systems, thus preventing accidents, improving sleep hygiene through pattern tracking (e.g., reducing nighttime awakenings), and enabling reduced staffing hours; this fosters greater independence, better health outcomes like fewer falls (studies show up to 50% reduction), and data logs that support personalized care plans.

Various vendors (together with cost): Simply Home ($100-200, simply-home.com, IDD integration), Smart Caregiver ($30, smartcaregiver.com, basic wireless), Rest Assured ($150+, rest-assured.com, telecare/fall features), and VitalBase VB-100 ($120, vitalbase.com, AI pattern learning).

4.4 Safety

Safety-focused technologies in smart homes prioritize hazard detection and prevention, encompassing sensors for doors, stoves, water leaks, and automated locks to create secure yet non-restrictive environments for IDD residents. These devices provide real-time alerts and automated responses to mitigate risks like fires, floods, or unauthorized access, integrating seamlessly with broader smart systems. People with IDD encounter significant safety challenges, such as difficulty perceiving dangers (e.g., forgetting appliances on due to attention deficits), vulnerability to accidents from impaired judgment, or elopement risks in unfamiliar settings, which can lead to injuries, property damage, or exploitation, straining caregiver resources and support systems.

4.4.1 Door Sensor

Functionality: Door sensors are wireless contact devices typically consisting of two parts—a magnet and a sensor—that detect when a door or window opens or closes. Placed on the frame and door itself, they trigger alerts via a connected hub or app when separation occurs, such as during unusual times like nighttime wandering. They integrate with smart home systems to activate lights, send notifications to caregivers, or play verbal prompts, using technologies like Zigbee or Wi-Fi for reliable, low-power operation.

How it helps IDD: Individuals with intellectual and developmental disabilities (IDD) often face challenges like wandering or elopement, which can lead to safety risks such as getting lost or exposure to hazards, especially during unsupervised hours in group homes or independent living

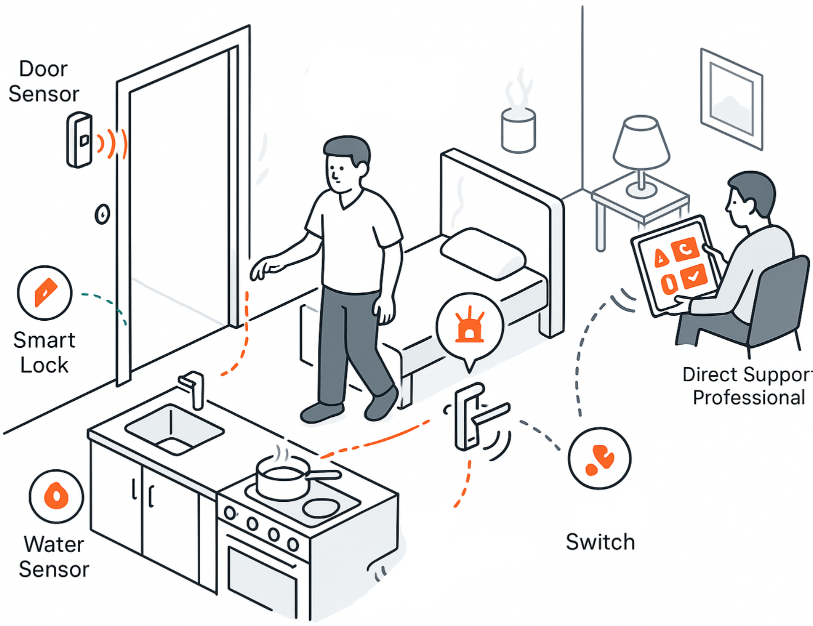

Figure 4.3: *Examples of safety-related sensors and devices in a smart home for IDD support.*

settings. Door sensors address this by providing real-time alerts to caregivers or remote support staff when a door opens unexpectedly, enabling quick interventions like verbal reminders or remote locking, thus promoting independence while reducing the need for constant physical supervision and lowering incident rates by up to 20-30% according to studies on remote monitoring for IDD populations.

Various vendors (together with cost): Aqara ($18, amazon.com, hub required), Simply Home ($50-100, simply-home.com, bundled with system), and YoLink ($20, shop.yosmart.com, long-range).

4.4.2 Stove Sensor

Functionality: A stove sensor is a smart device designed to monitor the usage and status of a stovetop, typically detecting whether the stove is on or off through current sensing or motion/heat detection without directly measuring temperature. It is usually installed directly on the stove's power outlet, under the cooktop, or nearby on the wall, and connects wirelessly to a central hub or app. When the stove is activated, the sensor tracks duration and patterns; if it detects prolonged use (e.g., beyond a customizable threshold like 30-60 minutes) or unusual activity (such as being left on unattended), it triggers real-time alerts via smartphone notifications, audible alarms, or automated responses like cutting power to the outlet if integrated with a smart switch, helping prevent fire hazards.

How it helps IDD: Stove sensors help prevent fire risks for people with IDD by alerting caregivers if the stove is left on, allowing for quick reminders or automatic shut-off. This improves safety and lets residents cook more independently with less need for constant supervision.

Various vendors (together with cost): Simply Home ($150-250, simply-home.com, IDD bundles), Innohome Stove Guard SGK510 ($399, innohome.com, auto shut-off), FireAvert ($149.99, fireavert.com, smoke alarm integration), and iGuardStove ($495, iguardstove.com, motion-based shut-off).

4.4.3 Water Sensor

Functionality: A water sensor, also known as a leak detector, is a compact battery-powered or wired device that detects the presence of water or moisture through conductive probes or optical sensors, typically placed on the floor near potential leak sources like sinks, toilets, washing machines, water heaters, or basements. It reacts instantly upon contact with water by sending wireless signals to a connected hub or app, triggering smartphone alerts, audible sirens (often 80-100 dB), or integrations with other smart devices like shut-off valves to automatically stop water flow; some models also monitor for freezing temperatures or humidity levels, providing early warnings to prevent minor leaks from escalating into floods.
How it helps IDD: People with IDD may struggle with self-care tasks, leading to common problems like forgetting to turn off faucets, overflows during bathing, or unnoticed leaks from appliances. The water sensor mitigates these by offering real-time detection and notifications to residents (via simple audio cues) or remote caregivers, enabling quick responses.
Various vendors (together with cost): YoLink ($18, yolink.com, hub required, loud alarm), Aqara ($19.99, aqara.com, Zigbee/app), Honeywell Lyric ($79, honeywellhome.com, Wi-Fi, cable extension), and Kidde ($39.99, kidde.com, battery, freeze detection).

4.4.4 Automated Door Lock/Opener

Functionality: An automated door lock/opener combines a smart deadbolt or handle with motorized mechanisms to lock/unlock and physically open/close doors, often installed on exterior or interior doors like front entrances, bedrooms, or bathrooms; it reacts to inputs such as app commands, voice controls, keypads, fingerprints, proximity sensors (e.g., geofencing via Bluetooth/GPS), or motion detection, automatically engaging locks for security or swinging doors open with low-energy actuators to comply with ADA standards, while integrating with cameras for visitor verification and sending status alerts.
How it helps IDD: Residents with IDD frequently encounter barriers to mobility and safety, such as difficulty manipulating keys due to fine motor skill impairments, forgetting to lock doors leading to vulnerability, or challenges entering/exiting during wandering episodes, which can result in elopement risks or unauthorized access. This device aids by enabling hands-free or remote operation—e.g., voice-activated unlocking via Alexa or automatic opening upon approach—fostering independence in daily routines like greeting visitors or accessing rooms, while caregivers receive alerts for unusual activity to prevent hazards.
Various vendors (together with cost): Schlage Encode ($224, schlage.com, app/voice, alarm), OlideSmart ($849, olidesmart.com, ADA opener), August Wi-Fi Smart Lock ($229.99, august.com, retrofit, auto-unlock), and Sesame by Candy House ($149.99+$99, candyhouse.co, opener add-on).

4.4.5 Outlet Switch/Light Switch

Functionality: An outlet switch (smart plug) or light switch (in-wall dimmer/toggle) is a Wi-Fi-enabled device that controls power to plugged-in appliances or wired lights, installed by plugging into standard outlets for switches or replacing existing wall switches; it reacts to app schedules, voice commands, motion sensors, or timers by turning devices on/off, dimming lights, or monitoring energy usage, with features like away modes that simulate occupancy by randomly activating connected items.
How it helps IDD: IDD individuals may forget to turn off lights/appliances leading to high energy costs, safety issues like overheating, or disrupted sleep from improper lighting. These switches assist by automating routines—e.g., scheduled light dimming for bedtime or voice-controlled appliance shut-off—empowering users to maintain independence with much less cognitive or physical effort.

Various vendors (together with cost): TP-Link Kasa ($44/4-pack plugs or $44 dimmer, tp-link.com, energy monitoring), Leviton Decora Smart ($49.99, leviton.com, in-wall/app), Brilliant ($399, brilliant.tech, all-in-one panel), and Enabling Devices ($29.95, enablingdevices.com, adapted/big-button).

4.5 Staff Communication

Staff communication technologies facilitate emergency response and ongoing interaction between IDD residents, caregivers, and support teams, using wearables like panic pendants and video doorbells to enable quick alerts, two-way dialogue, and remote monitoring. These tools bridge gaps in verbal or physical communication, ensuring timely assistance without invasive oversight. IDD individuals often struggle with communication due to speech impairments, anxiety in expressing needs, or delays in recognizing emergencies, leading to unmet support, isolation during crises, or over-reliance on in-person staff that can hinder independence and increase burnout.

4.5.1 Panic Pendant

Functionality: A panic pendant is a wearable emergency alert device, typically designed as a necklace or wristband with a large, easy-to-press button that, when activated, sends an immediate distress signal to predefined contacts, a monitoring center, or emergency services. It reacts to user-initiated presses or, in advanced models, automatic fall detection via built-in accelerometers, triggering alerts through cellular, Wi-Fi, or landline connections, often accompanied by two-way audio communication for verification. These devices are placed on the user's person for constant accessibility, such as around the neck for quick reach during daily activities or on the wrist for those with mobility preferences, ensuring they remain unobtrusive yet readily available in homes, outdoors, or care facilities.
How it helps IDD: Individuals with intellectual and developmental disabilities (IDD) often face heightened risks in emergencies due to challenges in communication, cognitive processing delays, or physical limitations that make it difficult to seek help independently. A panic pendant addresses these issues by providing a simple, one-touch mechanism for summoning assistance.
Various vendors (together with cost): Life Alert ($49.95/mo + $197, lifealert.com, 24/7 monitoring, contract), Assistive Technology Services ($119.99, assistivetechnologyservices.com, no monthly fee, basic), Medical Guardian Mini Guardian ($39.95/mo + $124.95, medicalguardian.com, GPS/mobile), and Bay Alarm Medical ($29.95/mo, bayalarmmedical.com, waterproof, home integration).

4.5.2 Ring Video Doorbell

Functionality: The Ring Video Doorbell is a smart home security device equipped with a high-definition camera, motion sensors, and two-way audio, which detects movement or doorbell presses and sends real-time notifications to a connected smartphone app, allowing users to view live video feeds, communicate with visitors, and record events. It reacts to customizable motion zones by triggering alerts. Typically placed at the front door at eye level for optimal coverage, it operates via Wi-Fi and can be hardwired or battery-powered, making it suitable for homes, apartments, or care facilities.
How it helps IDD: Residents with intellectual and developmental disabilities (IDD) frequently encounter safety concerns related to strangers at the door. The Ring Video Doorbell mitigates these by enabling remote monitoring and interaction, allowing IDD individuals or their caregivers to screen visitors via app alerts without opening the door.
Various vendors (together with cost): Ring ($49.99+ for Wired, $149.99 for Battery Plus, ring.com, Amazon, subscription for storage), Google Nest Doorbell ($179.99, store.google.com,

AI detection), Arlo Essential ($79.99, arlo.com, local storage), and Eufy ($99.99+, eufy.com, no monthly fee, privacy focus).

5 Implementation Guide

Author: Olzhas Yessenbayev

5.1 Vendor Selection: Simply Home

The exploration prioritizes offerings from **Simply Home** (simply-home.com), a technology provider which designs integrated systems to foster independence. Simply Home offers a range of sensing and assistive technologies which address key challenges such as wandering, medication non-adherence, cooking safety risks, and falls. Simply Home provides customizable assistive technology solutions that enable independent living, with case studies highlighting integrated systems involving sensors and monitors, providing real-time support to Direct Support Professionals (DSPs) and promoting resident autonomy.

5.1.1 Platform Suitability for IDD Population

SimplyHome's technologies are not exclusively tailored to IDD but are consumer-centered for disabilities, aging, and veterans, emphasizing independence via customizable alerts and prompts. Many features (e.g., verbal prompts, activity logging) align well with IDD needs like routine support and safety monitoring. However, some features (e.g., Environmental Controls for remote door unlocking via tablet) target low mobility, which could pose risks for IDD residents (e.g., accidental unlocking leading to wandering or unauthorized access).
This requires careful implementation with appropriate safeguards: For IDD populations, Direct Support Professionals (DSPs) or family members primarily control advanced features via the Responder App, while residents use simplified interfaces such as large-button tablets or voice prompts currently in development.

5.2 System Architecture and Integration

5.2.1 Device Ecosystem Overview

Devices are sold both separately (e.g., individual sensors) and in bundles (e.g., within the SimplyHome System models like Firefly or Butler, which include a base hub and custom sensors). Bundles are customized via assessments, with leasing options available. All devices connect through a central hub (Firefly or Butler) to SimplyHome's cloud service, enabling real-time data logging, alerts (text/email/phone), and remote management via the Responder App or web portal.

5.2.2 Communication and Data Flow

Communications are handled securely in the cloud, with options for internet or cellular connectivity. For example, in a CFS home: A resident's door sensor detects opening after 10 PM (trigger), sending an alert to a caregiver's app; the caregiver views the Ring camera feed remotely and issues a verbal prompt via a tablet in the home, reducing the need for on-site staff and saving $20-40/hour in labor costs per incident.

5.3 Implementation Process for CFS Family Homes

5.3.1 Assessment Phase

CFS Care Manager identifies resident goals (e.g., reduce wandering) via Life Plan. Refer to SimplyHome for assessment; customize bundle (e.g., door/motion sensors + hub).

5.3.2 Installation Process

SimplyHome technician installs (fee approximately $100-200, relocatable). Sensors placed unobtrusively (e.g., door sensors on exits, bed pads under mattresses). Hub connects to internet/-cellular; tablets mounted accessibly but securely.

5.3.3 Management Roles and Responsibilities

Residents

Use simple features (e.g., press panic button, respond to prompts like "Take meds"). For IDD, focus on passive monitoring to avoid overwhelm.

DSPs/Caregivers

Primary responders via Responder App. Receive alerts, view portal data, adjust rules (e.g., Firefly allows real-time changes). Document visits/check-ins for accountability.

Family/CFS Staff

Access web portal for trends (e.g., sleep reports to adjust care). 24/7 Response Center as backup for urgent alerts.

SimplyHome Support

Provides training, ongoing support, and privacy-balanced monitoring (no constant cameras; optional entry-way only).

5.4 Cost Structure

Bundles (e.g., SimplyHome System: Firefly ($849.95-1,149.90) or Butler ($1,249.90-1,549.90) + monthly fee ($79.95-$149.95) include hub + custom sensors (3-5 typically). Separate sales allow add-ons. All systems envision remote support: Sensors feed data to cloud; alerts/responses via app/portal, minimizing on-site presence.

5.5 Ethical Considerations and Safeguards

Ethical considerations, including robust data privacy protocols (e.g., encrypted transmissions and user consent for monitoring) and collaborative involvement of DSPs in setup and oversight, are woven into the evaluation to ensure technologies enhance care without compromising dignity or autonomy.

6
Conclusion

Author: Olzhas Yessenbayev

6.1 Value Recap

Tech justifies investment by enhancing lives, efficiency, and dignity.

6.2 Handoff Emphasis

Report as "fluid template" for interns (e.g., adaptable guide).

6.3 Final Justification

Align with CFS/PiTech missions; end with forward-looking statement.

Appendices

A
Showcasing the First Appendix

Writing Guidance

Appendices contain supplementary material **created by the author** that enhances the reader's understanding of the dissertation while not being essential for following the primary narrative. These sections often include detailed tables, figures, complex calculations, or materials like survey questions and interview transcripts produced in the course of the research. The appendices allow readers to explore the research in greater detail, offering a deeper insight into methods and findings without interrupting the main body of work.

B
Showcasing the Second Appendix

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Annexes

L
Showcasing the First Annex

Writing Guidance

Annexes are supplementary sections in a dissertation that provide additional information or external documents not essential to the main arguments but that support or complement the research. Unlike appendices, **annexes generally contain material that was not developed by the author**, such as reports, legal documents, or published datasets from external sources. This information is placed separately to keep the main content concise, allowing readers access to relevant external references without disrupting the dissertation's flow.

This page intentionally left blank.